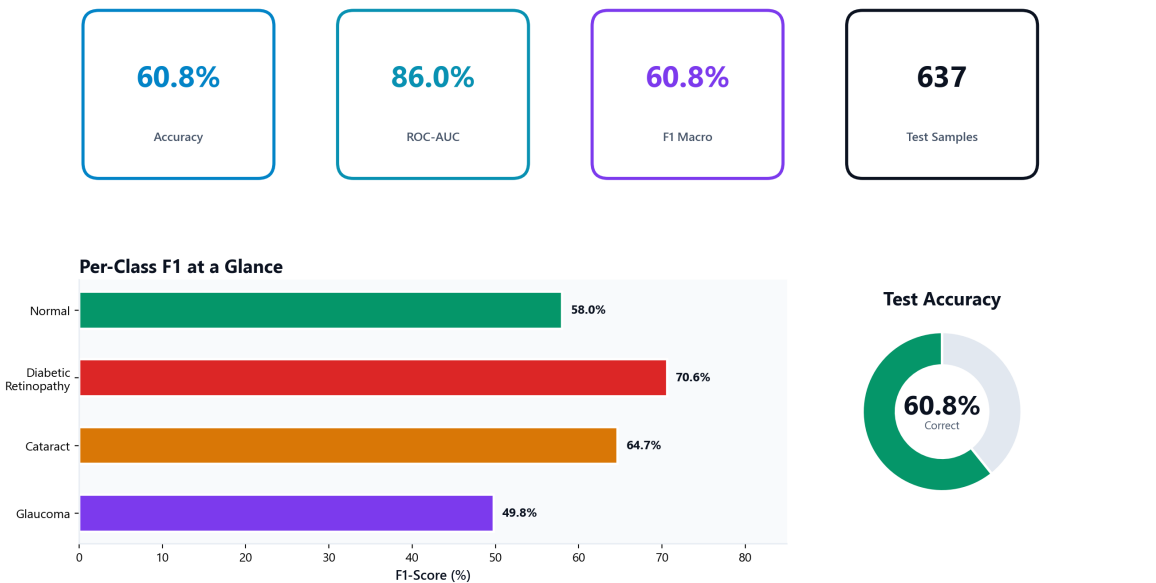


Explainable and Deployable Artificial Intelligence for Retinal Disease Classification

Document Type	Original Research — Parts 4 & 5
Focus	Explainable AI (Grad-CAM) · Deployable AI (Streamlit)
Modules	src/gradcam.py · src/app.py · src/report_pdf.py
Framework	TensorFlow/Keras · Streamlit · ReportLab
Date	June 09, 2026

Executive Summary — Explainable & Deployable AI (Parts 4 & 5)



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Figure 0. Executive summary of model performance and per-class F1 scores.

Abstract

Deep learning classifiers for retinal fundus images achieve useful discrimination but remain difficult to trust in clinical settings without transparent reasoning and accessible deployment. We present Parts 4 and 5 of an eye-disease classification system covering **Explainable AI** via Gradient-weighted Class Activation Mapping (Grad-CAM) and **Deployable AI** via a Streamlit web interface. Grad-CAM highlights spatial evidence from the final convolutional layer using TensorFlow GradientTape. The deployed application supports drag-and-drop upload, real-time inference, interactive heatmaps, and downloadable PDF reports. On a held-out test set ($n = 637$), the best model achieved 60.8% accuracy and 86.0% macro ROC-AUC. Grad-CAM enables human-in-the-loop verification, particularly for low-recall classes such as glaucoma (38.2% recall). This work demonstrates a reproducible pathway from model output to explainable, user-facing decision support for research and education.

Keywords: Explainable AI · Grad-CAM · Streamlit · Retinal Fundus · Clinical Decision Support · Deep Learning · Deployable AI

1. Introduction

Automated analysis of retinal fundus photographs supports early detection of diabetic retinopathy, cataract, and glaucoma. Convolutional neural networks and transfer-learning architectures such as ResNet50 and EfficientNetB0 provide strong feature extraction, yet two barriers limit translational use: models do not explain their decisions, and trained weights are not accessible to non-programmers.

Parts 4 and 5 address these gaps. Part 4 implements post-hoc explainability with Grad-CAM. Part 5 deploys inference and explanation through a Streamlit application with PDF export. We evaluate explainability in the context of quantitative test-set performance and discuss limitations relevant to responsible medical AI research.

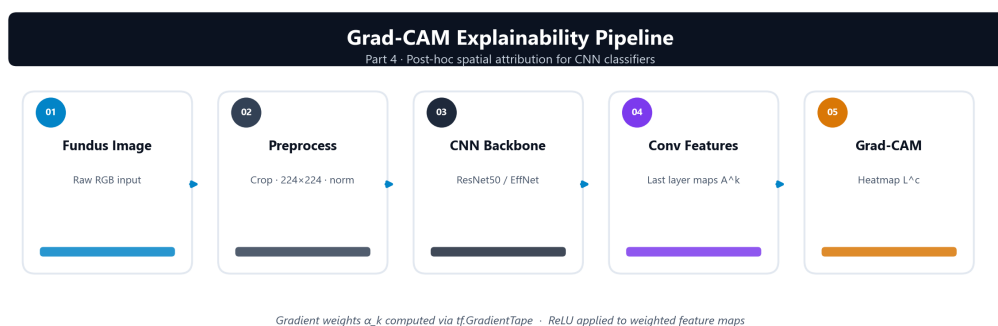
2. Methods

2.1 Dataset and Model Context

We used the Kaggle Eye Diseases Classification dataset with four classes: Normal, Diabetic Retinopathy, Cataract, and Glaucoma. Images were preprocessed (medical crop, 224×224 resize, normalization) as described in Parts 1–3. The best saved model was evaluated on 637 test images with stratified class representation.

2.2 Grad-CAM Explainability (Part 4)

We implemented Grad-CAM in `src/gradcam.py`. Given final convolutional feature maps A^k and class score y^c , importance weights are computed by global average pooling of gradients $\partial y^c / \partial A^k$. The localization map $L = \text{ReLU}(\sum_k \alpha_k A^k)$ is upsampled and alpha-blended ($\alpha = 0.4$) onto the original image. The last convolutional layer is auto-detected for Keras 2/3 compatibility.

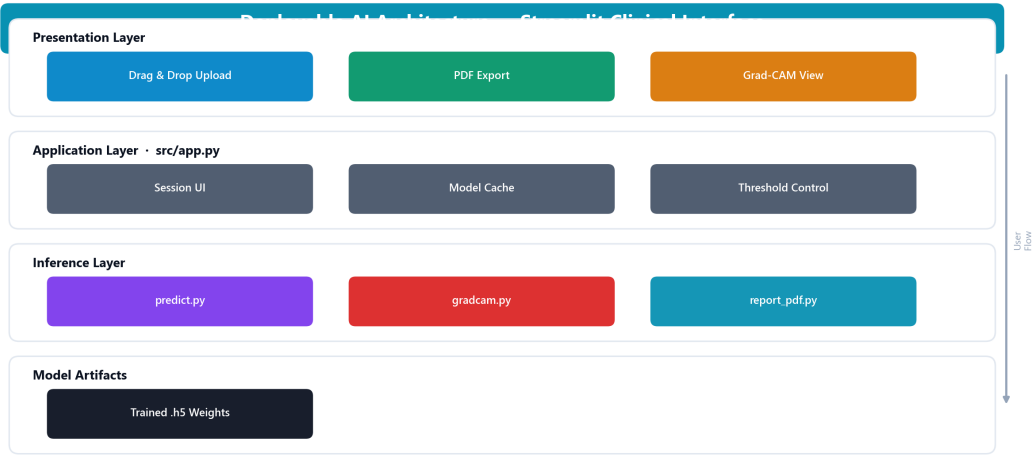


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Figure 1. Grad-CAM explainability pipeline from fundus input to spatial heatmap.

2.3 Streamlit Deployment (Part 5)

We built a Streamlit interface in `src/app.py`. The application loads cached Keras models, accepts drag-and-drop uploads, runs `EyeDiseasePredictor`, optionally generates Grad-CAM overlays, and exports session reports as PDF via `src/report_pdf.py`. A configurable confidence threshold triggers low-confidence warnings.



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Figure 2. Layered deployment architecture for the Streamlit clinical interface.

Component	Module	Function
Grad-CAM engine	gradcam.py	Heatmap computation and overlay
Predictor	predict.py	Preprocessing and softmax inference
Web interface	app.py	Upload, visualization, export
PDF generator	report_pdf.py	Session report documentation

3. Results

3.1 Quantitative Performance

Table 1 summarizes test-set performance. Accuracy was 60.8%, macro ROC-AUC was 86.0%, and macro F1 was 60.8%. ROC-AUC exceeded accuracy, indicating separable class probabilities despite moderate hard-label accuracy.

Metric	Value
Accuracy	60.75%
Precision (macro)	68.82%
Recall (macro)	60.44%
F1-Score (macro)	60.77%
ROC-AUC (macro)	86.04%
Test samples	637

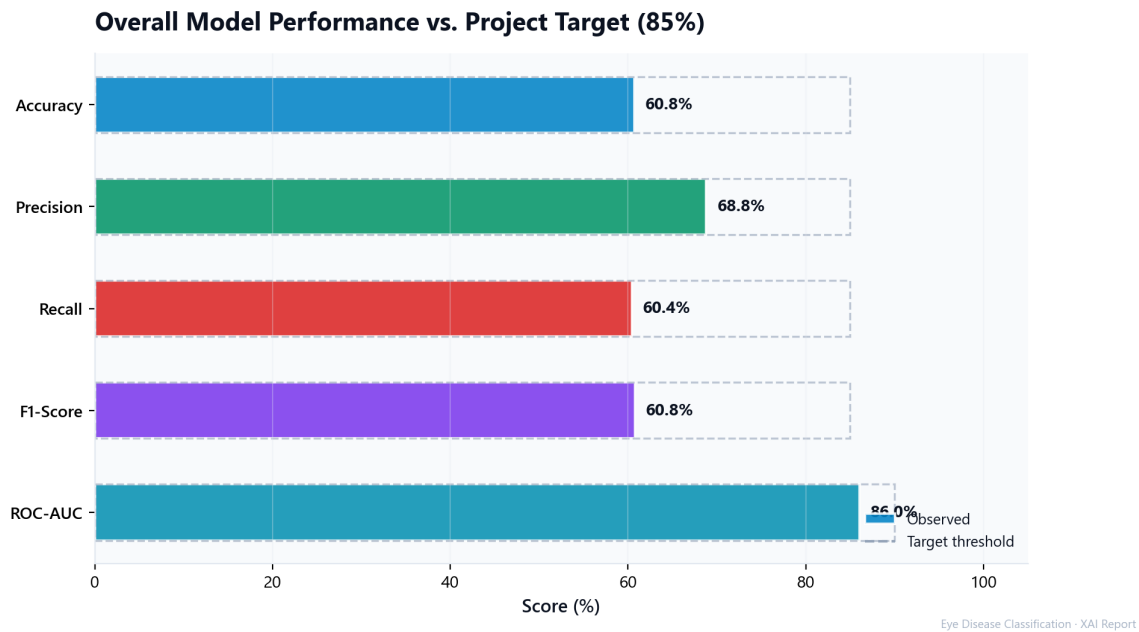


Figure 3. Overall metrics compared with the 85% project target.

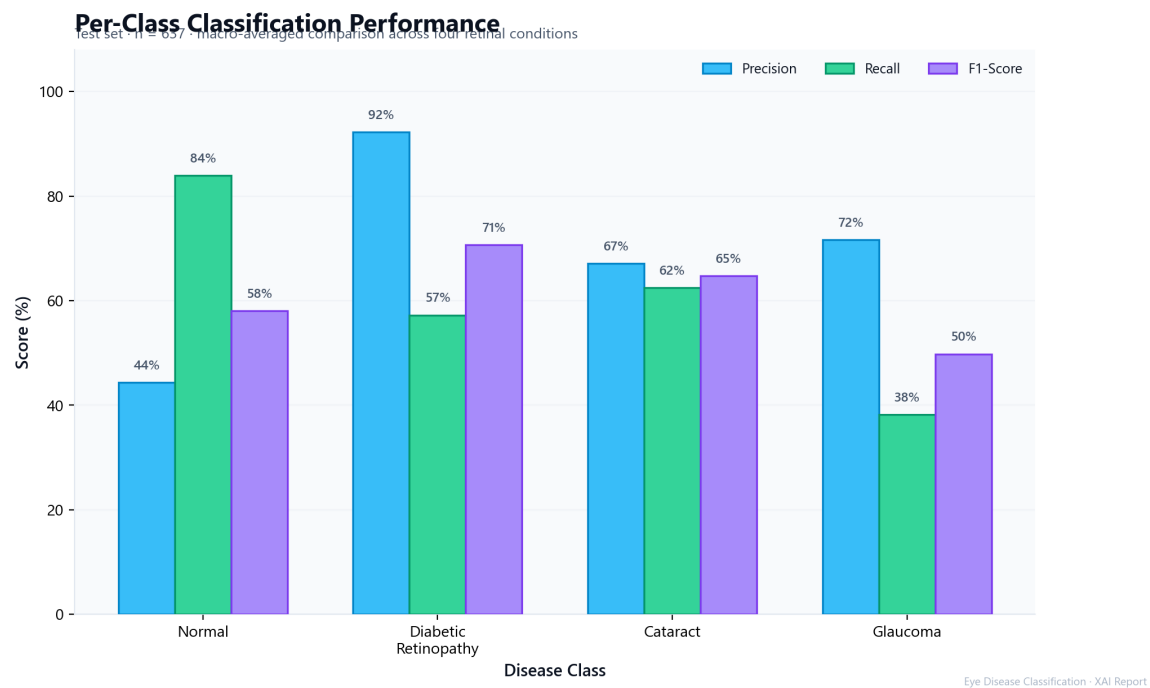


Figure 4. Per-class precision, recall, and F1 on the test set.

Confusion Matrix Analysis — Best Model

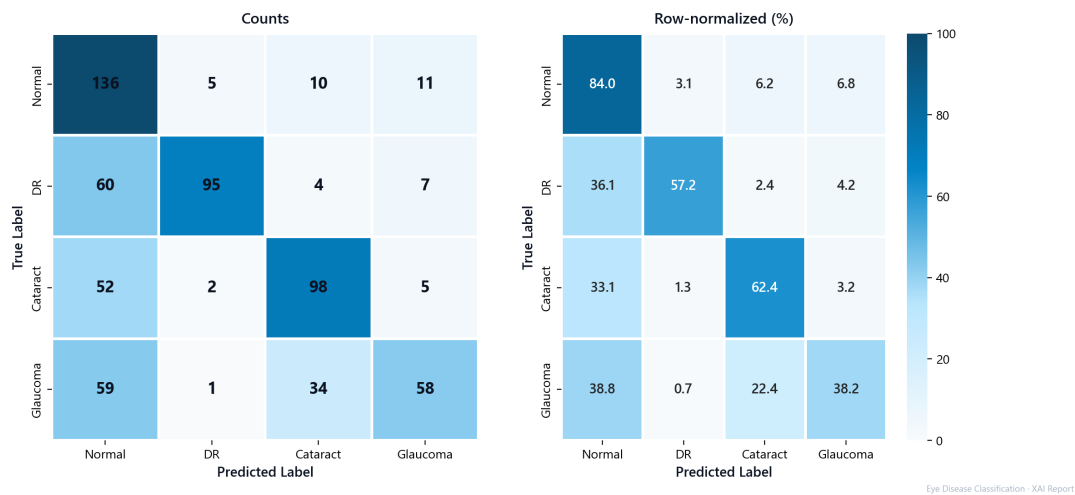


Figure 5. Confusion matrix — counts and row-normalized percentages.

3.2 Explainability Outputs

Grad-CAM produced three-panel visualizations: original fundus, raw heatmap, and clinical overlay. Attention concentrated on the optic disc and peripapillary region — anatomically relevant for glaucoma and vascular pathology. When heatmaps highlight acquisition borders, this indicates preprocessing or dataset artifacts rather than pathology.

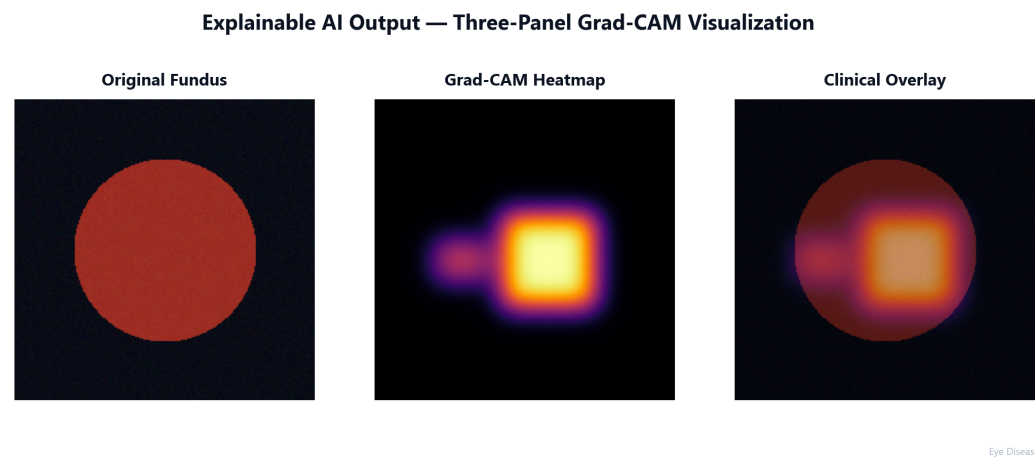


Figure 6. Representative Grad-CAM output panels (demonstration layout).

Class	Precision	Recall	F1	Support
Normal	44.3%	84.0%	58.0%	162
Diabetic Retinopathy	92.2%	57.2%	70.6%	166
Cataract	67.1%	62.4%	64.7%	157
Glaucoma	71.6%	38.2%	49.8%	152

4. Discussion

Integrating Grad-CAM with Streamlit creates a human-in-the-loop workflow: users receive a quantitative prediction and immediately inspect spatial evidence. This closed loop supports trust calibration — essential when accuracy remains below the 85% clinical target. Diabetic retinopathy achieved 92.2% precision but only 57.2% recall; glaucoma recall was 38.2%. Grad-CAM review is therefore most valuable for suspected false negatives in glaucoma screening.

4.1 Limitations

- Test accuracy (60.8%) limits clinical applicability; deployment is research-only.
- Grad-CAM layer auto-detection may fail on some Keras 3 topologies.
- Single-modality fundus input — no OCT or patient metadata.
- Local deployment without authentication or HIPAA-compliant infrastructure.

4.2 Future Work

We plan to evaluate Grad-CAM++ and Score-CAM, improve glaucoma recall via class-weighted loss, add batch screening with CSV export, and containerize deployment with Docker or Streamlit Cloud for reproducible demonstrations.

5. Conclusion

Parts 4 and 5 complete the translational pathway from trained classifier to explainable, deployable decision support. Grad-CAM reveals spatial attribution aligned with retinal anatomy; the Streamlit application delivers accessible inference, visualization, and PDF documentation. Together, these components demonstrate responsible, transparent AI design for retinal disease screening research.

Medical Disclaimer. This system is for educational and research purposes only. It does not replace professional medical diagnosis or treatment.

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